Family Name ....

Given Name ....

Student No. ....

Signature ....

THE UNIVERSITY OF NEW SOUTH WALES

School of Electrical Engineering & Telecommunications

FINAL EXAMINATION

Term 1, 2020

ELEC1111
Electrical and Telecommunications Engineering

TIME ALLOWED:	**2 hours**
TOTAL MARKS:	**100**
TOTAL NUMBER OF QUESTIONS:	**5**

THIS EXAM CONTRIBUTES 50% TO THE TOTAL COURSE ASSESSMENT

Reading Time: 10 minutes.

This paper contains 6 pages.

Candidates must **<u>ATTEMPT ALL</u>** questions.

Answer each question in a **separate answer booklet**.

Marks for each question are indicated beside the question.

This paper **<u>MAY NOT</u>** be retained by the candidate.

Print your name, student ID and question number on the front page of each answer book.

Authorised examination materials:

Candidates should use their own UNSW-approved electronic calculators.

This is a closed book examination.

Assumptions made in answering the questions should be stated explicitly.

All answers must be written in ink. Except where they are expressly required, pencils **may only be used** for drawing, sketching or graphical work.

QUESTION 1 [20 marks]

a. (10 marks) For the circuit shown in Figure 1,

i. (7 marks) Apply mesh analysis and write down the mesh equations using the labels provided. Simplify the equations in the form $ai_1 + bi_2 + ci_3 = d$, where coefficients a, b, c and d can be positive or negative.

Note: DO NOT solve the equations.

ii. (3 marks) Given the values of mesh currents as $i_1 = 3.5\,A$ $i_2 = -0.5\,A$ and $i_3 = 2.5\,A$ find the power in the dependent voltage source.

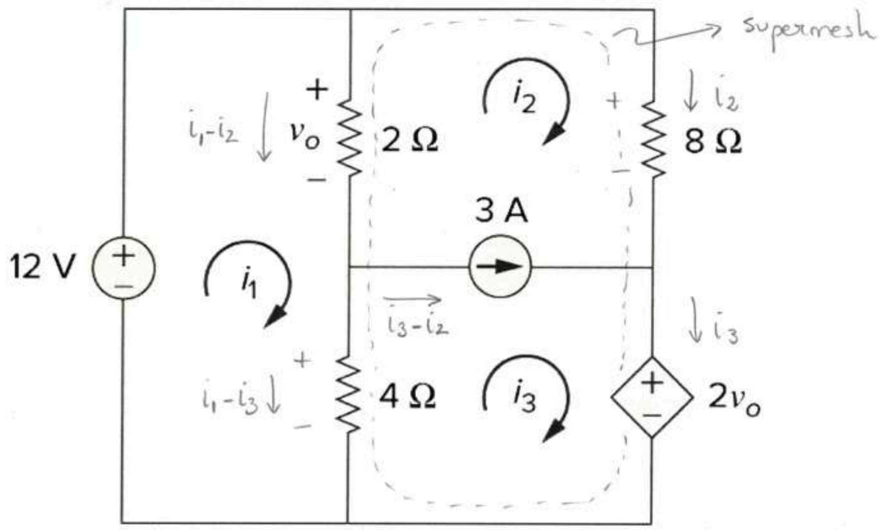


Figure 1

i)

KVL @ mesh 1 : $-12 + 2(i_1 - i_2) + 4(i_1 - i_3) = 0$ ①

KVL @ supermesh: $-4(i_1 - i_3) - 2(i_1 - i_2) + 8i_2 + 2v_o = 0$ ②, where $v_o = 2(i_1 - i_2)$ ④

Constraint equation: $i_3 - i_2 = 3$ ③

We simplify the equations :

$$6i_1 - 2i_2 - 4i_3 = 12 \quad ①$$

$$-4(i_1 - i_3) - 2(i_1 - i_2) + 8i_2 + 4(i_1 - i_2) = 0 \Rightarrow -2i_1 + 6i_2 + 4i_3 = 0 \quad ②$$

$$-i_2 + i_3 = 3 \quad ③$$

Putting them together:

$$\left.\begin{aligned} 6i_1 - 2i_2 - 4i_3 &= 12 \\ -2i_1 + 6i_2 + 4i_3 &= 0 \\ -i_2 + i_3 &= 3 \end{aligned}\right\}$$

ii)

$$\boxed{P_{ds} = + V_{ds} \times i_{ds} = 2v_o \times i_3 = 2 \times 2(i_1 - i_2) \times i_3 = 4 \times (3.5 + 0.5) \times 2.5 = 40\,W}$$

b. (10 marks) Transistors can serve as amplifiers, producing both current gain and voltage gain. They can be used to furnish a considerable amount of power to devices that convert energy, such as loudspeakers or control motors. Calculate the gain (i.e. v_0/v_s) of the transistor amplifier circuit shown in Figure 2 when $v_s = 2V$.

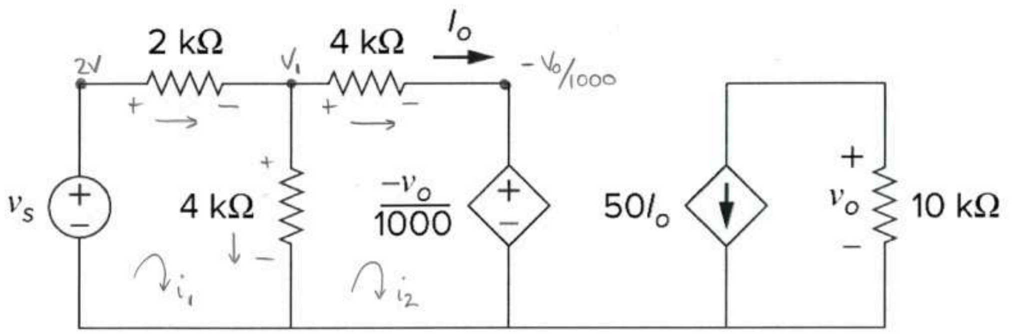


Figure 2

$V_o = -10K \times 50\, I_o$

We calculate I_o from the left hand side of the circuit:

* Using <u>mesh</u> analysis

$-2 + 2K i_1 + 4K(i_1 - i_2) = 0$

$-4K(i_1 - i_2) + 4K i_2 + \left(\frac{-V_o}{1000}\right) = 0$

where $i_2 = I_o$ and $V_o = -500K\, I_o$

We simplify the equations:

$6000\, i_1 - 4000\, I_o = 2$

$-4000\, i_1 + 8000\, I_o + \frac{500K\, I_o}{1000}$

$6000\, i_1 - 4000\, I_o = 2 \Rightarrow 6000\, i_1 - 4000\, I_o = 2$

$-40\, i_1 + 85\, I_o = 0 \xRightarrow{\times 150} -6000\, i_1 + 12750\, I_o = 0$

$8750\, I_o = 2$

$\underline{I_o = 228.57\ \mu A}$

From I_o we calculate V_o:

$V_o = -10K \times 50\, I_o = -0.5 \times 10^6\, I_o = -0.5 \times \cancel{10^6} \times 228.57 \times \cancel{10^{-6}} = \underline{-114.285\ V}$

Then, we calculate the gain:

$$\boxed{\frac{V_o}{V_s} = \frac{-114.285}{2} = -57.142}$$

* Using nodal analysis:

$$\frac{2-V_1}{2k} = \frac{V_1 - \left(\frac{-V_0}{1000}\right)}{4k} + \frac{V_1}{4k}$$

where $V_0 = -500k\, I_0$ and $I_0 = \frac{V_1 - \left(\frac{-V_0}{1000}\right)}{4k}$, which leads to $\frac{-V_0}{500\cancel{k}} = \frac{V_1 + \frac{V_0}{1000}}{4\cancel{k}}$

We simplify the equations:

$$\left.\begin{aligned} 4-2V_1 &= V_1 + \frac{V_0}{1000} + V_1 \\ -V_0 &= 125V_1 + \frac{125V_0}{1000} \end{aligned}\right\} \quad \left.\begin{aligned} \frac{V_0}{1000} &= 4-4V_1 \\ -1000V_0 &= 125000V_1 + 125V_0 \end{aligned}\right\} \quad \begin{aligned} 4000V_1 + V_0 &= 4000 \quad (\times(-1125)) \\ 125000V_1 + 1125V_0 &= 0 \end{aligned}$$

$$\Downarrow$$

$$\begin{aligned} -4500000V_1 - \cancel{1125V_0} &= -4500000 \\ 125000V_1 + \cancel{1125V_0} &= 0 \\ \hline -4375\cancel{000}V_1 &= -4500\cancel{000} \end{aligned}$$

$$\underline{V_1 = 1.0286\,V}$$

We calculate V_0 from V_1:

$$V_0 = 4000 - 4000V_1 = \underline{-114.4V}$$

Then, we calculate the gain:

$$\boxed{\frac{V_0}{V_s} = \frac{-114.4V}{2V} = \boxed{-57.2}}$$

QUESTION 2 [20 marks]

a. **(8 marks)** A magnetically controlled switch is called a *relay*. A relay is essentially an electromagnetic device used to open or close a switch that controls another circuit (see Figure 3(a)). The coil circuit is an RL circuit like that in Figure 3(b), where R and L are the resistance and inductance of the coil.

A certain relay has a resistance of 200Ω and an inductance of 500 mH. The relay contacts close when the current through the coil reaches 175 mA. What time elapses between the application of 110V to the coil (when S_1 closes at t = 0) and contact closure?

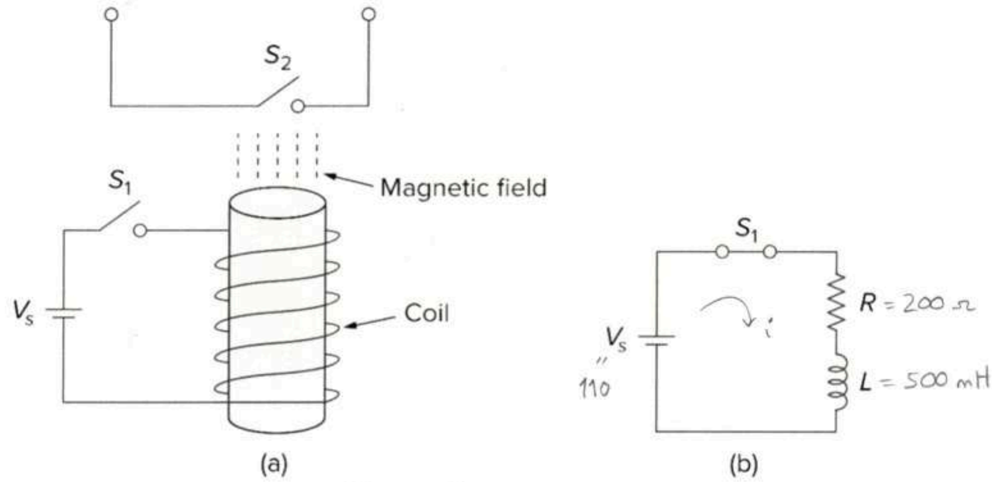


Figure 3

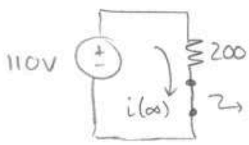
When S_1 closes at $t=0$ (i.e. $t>0$) $\Rightarrow$ step response

$$i(t) = i(\infty) + (i(0) - i(\infty))e^{-t/\tau}, \quad \text{where } \tau = L/R$$

* Calculate $i(0)$:

$i(0) = 0$ A

* Calculate $i(\infty)$:



$\rightarrow$ inductor is s.c. in steady state

$$i(\infty) = \frac{110}{200} = 0.55\ A$$

* Calculate τ:

$$\tau = L/R = \frac{500\ mH}{200\ \Omega} = 2.5\ ms$$

* Substituting in the equation: $i(t) = 0.55\left(1 - e^{-t/0.0025}\right)$ A

Contacts close when $i(t) = 175$ mA, so:

$$0.55\left(1 - e^{-400t}\right) = 0.175$$

$$+0.55e^{-400t} = +0.375$$

$$e^{-400t} = 0.6818$$

$$+400t = +0.383$$

$$\boxed{t = 957.5\ \mu s}$$

b. (12 marks) For the circuit shown in Figure 4,

i. (2.5 marks) Find the initial voltage $v(0^-)$ across the capacitor under steady-state condition.

ii. (2.5 marks) Find the final voltage $v(\infty)$ across the capacitor under steady-state condition.

iii. (4 marks) Derive an expression for the voltage of the capacitor $v(t)$ for all time (i.e., for both $t < 0$ and $t \geq 0$).

iv. (3 marks) Derive an expression for the current $i(t)$ for $t \geq 0$.

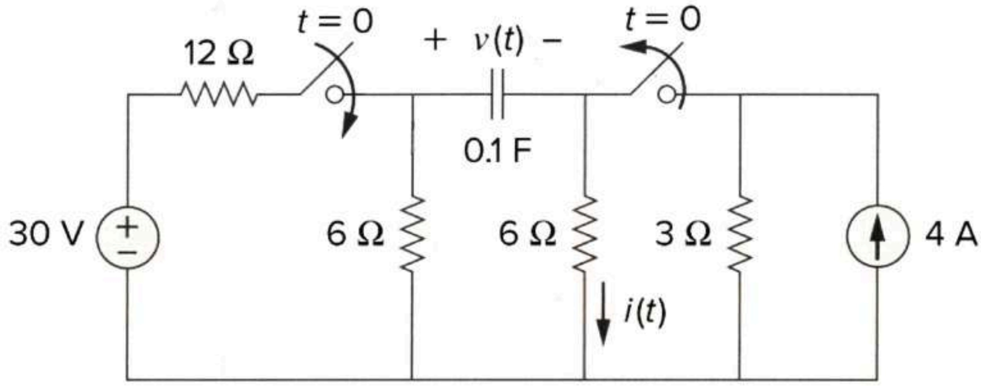
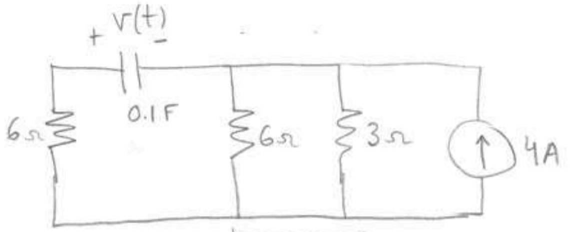


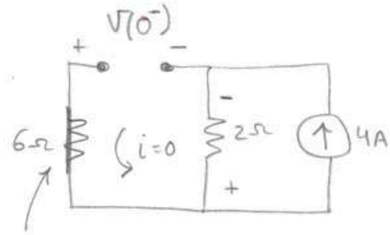
Figure 4

i) $t < 0$



$6 \| 3 = 2\Omega$

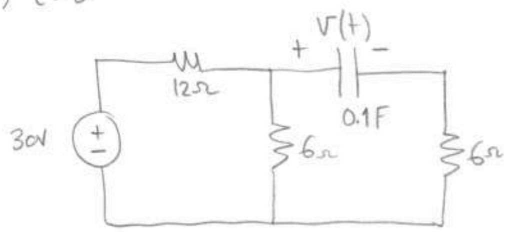
C ⇒ o.c. in DC steady state



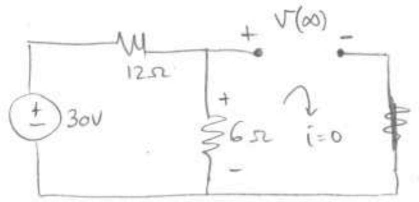
$V_{6\Omega} = 6\Omega \times 0A = 0V$

$\boxed{v(0^-) = V_{2\Omega} = 2\Omega \times (-4A) = -8V}$

ii) $t > 0$



C ⇒ o.c. in DC steady state



$(V_{6\Omega} = 0)$

$\boxed{v(\infty) = V_{6\Omega} = 30 \frac{6}{12+6} = 10\ V}$

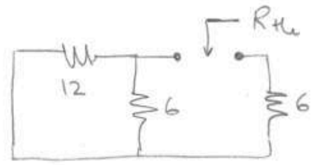
voltage divider

iii) For $t > 0$, step response

$v(t) = v(\infty) + (v(0) - v(\infty))\, e^{-t/\tau}$, where $\tau = R_{th} \times C$

$v(t) = 10 + (-8-10)\, e^{-t/\tau} = 10 - 18 e^{-t/\tau}$

We calculate τ from circuit @ $t > 0$



$R_{th} = (12 \| 6) + 6 = 4 + 6 = 10\Omega \Rightarrow \tau = R_{th} \times C = 10 \times 0.1 = 1s$

We substitute in the step response equation :

$$v(t) = 10-18\,e^{-t/\tau} = 10-18\,e^{-t} \text{ V} \quad \text{for } t \geq 0$$

Including the info for $t < 0$:

$$\begin{cases} v(t) = -8 \text{ V} & , t < 0 \\ v(t) = 10-18e^{-t} \text{ V} & , t \geq 0 \end{cases}$$

iv) $i = C\dfrac{dv}{dt}$

$$i(t) = 0.1 \times \frac{d}{dt}\left(10-18e^{-t}\right) = 0.1 \times (-18)\,e^{-t}/(-1) = 1.8\,e^{-t}$$

QUESTION 3 [25 marks]

a. **(12 marks)** In the circuit of Figure 5, calculate the voltage v_0 if $v_s = 2V$.

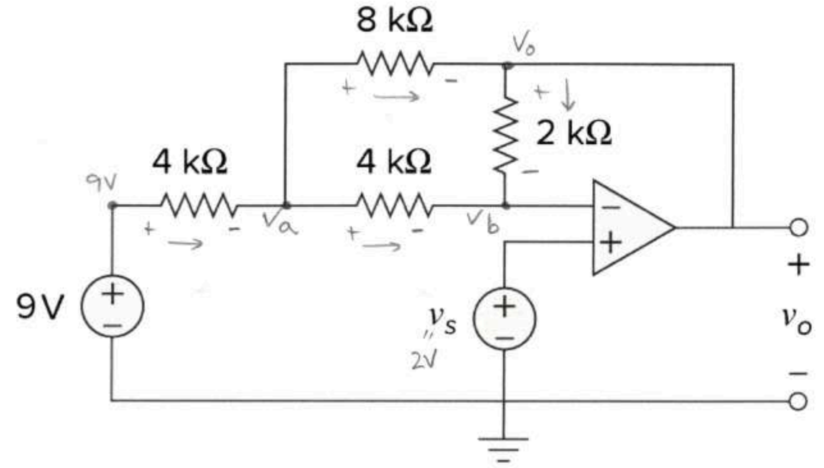


Figure 5

KCL@ node a:

$$\frac{9 - V_a}{4\cancel{k}} = \frac{V_a - V_o}{8\cancel{k}} + \frac{V_a - V_b}{4\cancel{k}} \quad ①$$

KCL@ node b:

$$\frac{V_a - V_b}{4\cancel{k}} + \frac{V_o - V_b}{2\cancel{k}} = 0 \quad ②$$

Also, $V_b = V^- = V^+ = 2V$

We substitute and simplify:

From ②:

$$\frac{V_a}{4} - \frac{1}{2} + \frac{V_o}{2} - 1 = 0 \; ; \quad V_a - 2 + 2V_o - 4 = 0$$

$$V_a = 6 - 2V_o$$

Substituting in ①:

$$9/4 - \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{4}\right)(6 - 2V_o) + \frac{V_o}{8} + \frac{1}{2} = 0$$

$\times 8$

$$18 - 5(6 - 2V_o) + V_o + 4 = 0$$

$$18 - 30 + 10V_o + V_o + 4 = 0$$

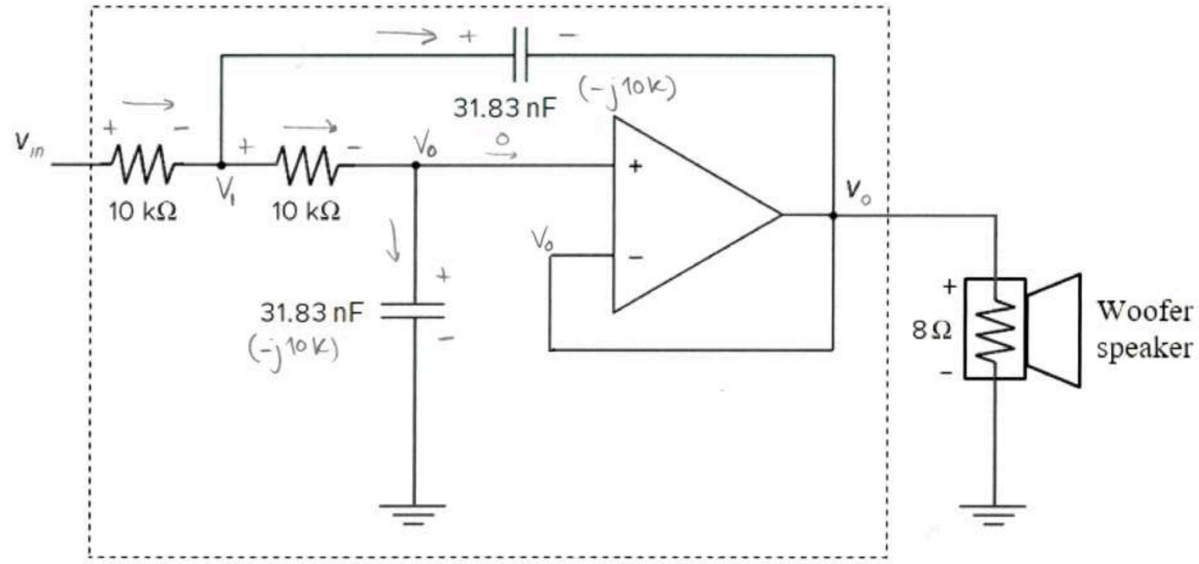
$$-8 + 11V_o = 0 \Rightarrow \boxed{V_o = 8/11 = 0.727\ V}$$

b. (13 marks) Woofers or bass speakers are used to reproduce low frequency sounds, typically from 40 Hz up to 500 Hz. The circuit in Figure 6 is used to set up a woofer speaker using a Sallen–Key low pass filter. If V_{in} is an AC signal with RMS value $V_{in_{RMS}}$ = 500 mV and frequency $f = 500\ Hz$, calculate:

i. **(2 marks)** Input voltage V_{in} as a function of time and then as a phasor.

ii. **(8 marks)** Output voltage V_0 of the Sallen-Key low pass filter.

iii. **(3 marks)** Average power absorbed by the woofer speaker (i.e. 8Ω resistor).



Sallen-Key low pass filter

Figure 6

i) $V_{in_{RMS}} = \frac{V_{in_m}}{\sqrt{2}} \Rightarrow V_{in_m} = \sqrt{2} \times 500 mV = 0.707\ V$

$V_{in} = 0.707 \cos(1000\pi f)\ V \Rightarrow V_{in} = 0.707 \angle 0°\ V$

ii) First, calculate the impedance of the capacitor:

$Z_{C_1} = Z_{C_2} = -j\frac{1}{\omega C} = -j\frac{1}{2\pi \times 500 \times 31.83 \times 10^{-9}} = -j\,10k$

Then, do nodal analysis to find V_o:

KCL @ node V_1: $\frac{V_{in} - V_1}{10k} = \frac{V_1 - V_o}{10k} + \frac{V_1 - V_o}{-j10k} \Rightarrow -jV_{in} + jV_1 = -jV_1 + jV_o + V_1 - V_o$

KCL @ node V_o: $\frac{V_1 - V_o}{10k} = \frac{V_o}{-j10k} \Rightarrow -jV_1 + jV_o = V_o$

Substitute V_{in} and simplify

$-j0.707 + (-1+2j)V_1 = (-1+j)V_o$

$-jV_1 = (1-j)V_o$

$-j0.707 + (-1+j)V_1 = 0$

$V_1 = \frac{j0.707}{-1+j} = \frac{0.707\angle 90°}{1.414\angle 135°} = 0.5\angle -45°$

$\boxed{V_o = \frac{-jV_1}{(1-j)} = \frac{0.5\angle -135°}{1.414\angle -45°} = 0.354\angle -90°\ V}$

iii)

$\boxed{P_{avg_{8\Omega}} = \frac{1}{2}\frac{|V_o|^2}{8\Omega} = \frac{1}{2}\frac{0.354^2}{8} = 7.83\ mW}$

QUESTION 4 [25 marks]

a. (10 marks) A phase shifting circuit is often employed to correct an undesirable phase shift already present in a circuit or to produce special desired effects. The circuit in Figure 7 shows an RC phase shift circuit with a certain lagging phase shift. Calculate the value of $\mathbf{V}_0$ as a function of $\mathbf{V}_i$ and state the phase shift introduced by the circuit.

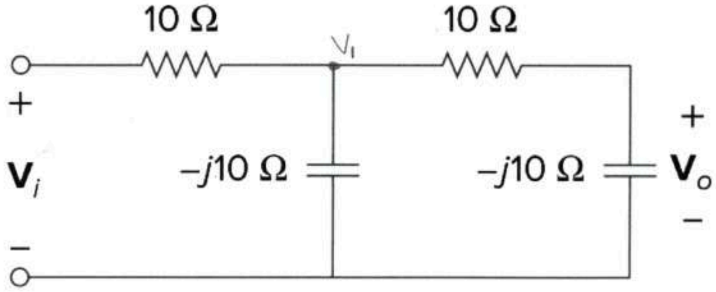
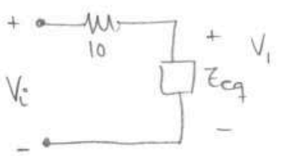


Figure 7

Calculate V_1 using voltage division:



$$V_1 = V_i \frac{Z_{eq}}{10+Z_{eq}} \quad \text{, where} \quad Z_{eq} = -j10 \parallel (10-j10)$$

$$Z_{eq} = \frac{-j1\cancel{0}(10-j10)}{-j1\cancel{0}+1\cancel{0}-j1\cancel{0}} = \frac{-j(10-j10)}{1-2j} = \frac{-10-j10}{1-j2} = \frac{14.142\angle -135^\circ}{2.23\angle -63.43} =$$

$$= 6.342\angle -71.57 = 2-j6$$

$$V_1 = V_i \frac{6.342\angle -71.57}{12-j6} = V_i \frac{6.342\angle -71.57}{13.42\angle -26.56} = 0.472\angle -45^\circ \times V_i$$

Then, calculate V_o from V_1 using voltage division again:

$$V_o = V_1 \frac{-j10}{10-j10} = V_i \times 0.472\angle -45^\circ \times \frac{10\angle -90^\circ}{10-j10} = V_i \times 0.472\angle -45^\circ \times \frac{10\angle -90^\circ}{14.14\angle -45^\circ} = 0.334\angle -90^\circ \times V_i$$

b. **(15 marks)** For the circuit shown in Figure 8,

i. **(5 marks)** Calculate the load impedance $\boldsymbol{Z}_L$ that will ensure the maximum transfer of power from the circuit to the load.

ii. **(10 marks)** Find the maximum average power received by $\boldsymbol{Z}_L$.

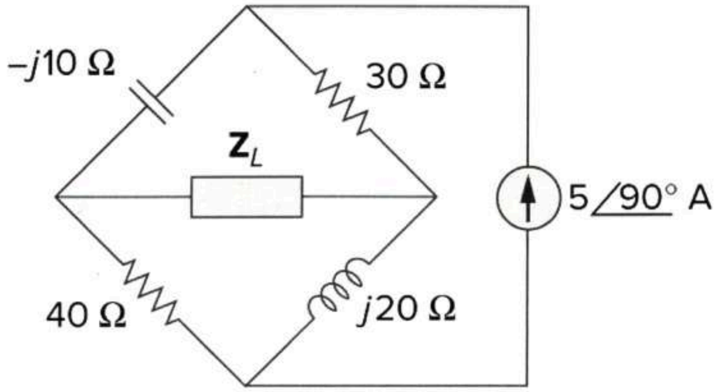
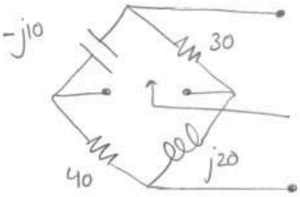


Figure 8

i) $Z_L = Z_{th}^*$

We turn off current source and calculate equivalent impedance to calculate Z_{th}:



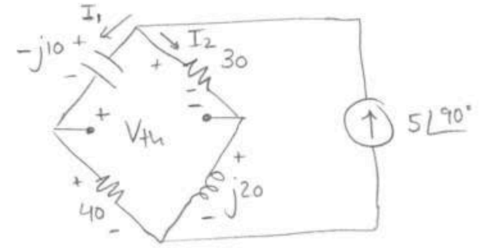
$$Z_{th} = (30-j10) \parallel (40+j20) = \frac{(30-j10)(40+j20)}{30-j10+40+j20} = \frac{120\cancel{0} + j60\cancel{0} - j40\cancel{0} + 20\cancel{0}}{7\cancel{0}+j2\cancel{0}}$$

$$= \frac{140+j20}{7+j} = 20\ \Omega$$

$$\boxed{Z_L = Z_{th}^* = 20\Omega}$$

ii) $P_{max} = \dfrac{|V_{th}|^2}{8 R_{th}}$

We calculate V_{th}:



Calculate I_1 and I_2 using current division:

$$I_1 = 5\angle 90^\circ \frac{30+j20}{30+j20+40-j10} = 5\angle 90^\circ \frac{30+j20}{70+j10} =$$

$$= -1.1 + j2.3 = 2.55\angle 115.56^\circ \text{ A}$$

$$I_2 = 5\angle 90^\circ \frac{40-j10}{70+j10} = 1.1 + j2.7 = 2.92\angle 67.83^\circ \text{ A}$$

Calculate V_{th} doing KVL in the upper mesh:

$$-j10 I_1 + V_{th} - 30 I_2 = 0 \quad \Rightarrow V_{th} = j10 I_1 + 30 I_2$$

$$V_{th} = 10\angle 90^\circ \times 2.55\angle 115.56 + 30 \times 2.92\angle 67.83^\circ = 25.5\angle -154.44 + 87.6\angle 67.83^\circ =$$

$$= (-23 - j11) + (33.06 + j81.12) = 10.06 + j70.12 = 70.84\angle 81.84^\circ$$

Then calculate P_{max}:

$$\boxed{P_{max} = \frac{70.84^2}{8 \times 20} = 31.36\text{W}}$$

IGNORE THIS QUESTION

QUESTION 5 [10 marks]

a. **(4 marks)** Simplify the following logical expression:

$$Z = \overline{A + \overline{B}C + \overline{C}}$$

b. **(6 marks)** Consider the logical diagram shown in Figure 9.

i. **(1.5 marks)** Derive the logical expression for Y.

ii. **(4.5 marks)** Construct the truth table relating Y to inputs A, B and C.

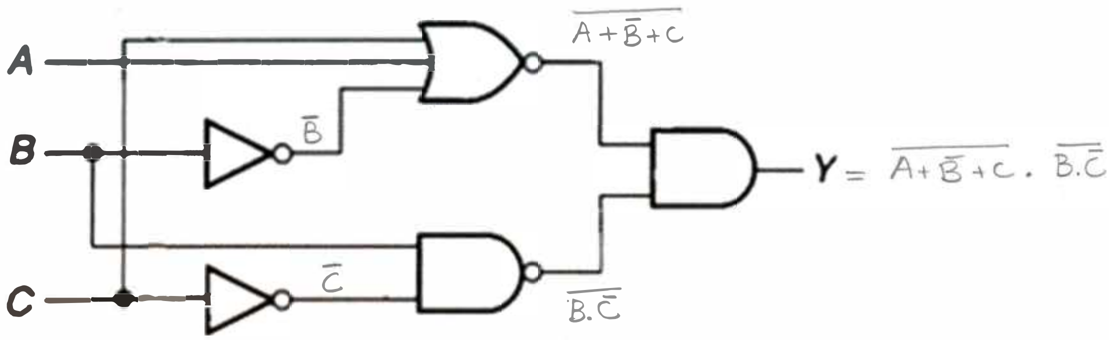


Figure 9

a)

$$Z = \overline{A+\overline{B}C+\overline{C}} = \overline{A}.\overline{\overline{B}C}.\overline{\overline{C}} = \overline{A}.(\overline{\overline{B}}+\overline{C}).C = \overline{A}.B.C + \overline{A}.\overline{C}.C = \overline{A}BC + 0 = \overline{A}BC$$

De Morgan's

De Morgan's $\overline{\overline{C}} = C$

Distributive law $\overline{\overline{B}} = B$

$\overline{C}C = 0$, $X.0 = 0$

$X + 0 = X$

b) i) $Y = \overline{A+\overline{B}+C} \cdot \overline{B.\overline{C}}$ (see diagram for intermediate steps)

ii)

A	B	C	$\overline{B}$	$\overline{C}$	$A+\overline{B}+C$	$\overline{A+\overline{B}+C}$	$B.\overline{C}$	$\overline{B.\overline{C}}$	Y
0	0	0	1	1	1	0	0	1	0
0	0	1	1	0	1	0	0	1	0
0	1	0	0	1	0	1	1	0	0
0	1	1	0	0	1	0	0	1	0
1	0	0	1	1	1	0	0	1	0
1	0	1	1	0	1	0	0	1	0
1	1	0	0	1	1	0	1	0	0
1	1	1	0	0	1	0	0	1	0